

Merge Sort

Core functions
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CODICE

procedure MERGE (A, l, mid, r)

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procedure MERGE_SORT (A, l, r)

if $l < r$ then

$mid = (l + r) / 2$] DIVIDE

MERGE_SORT (A, l, mid)]

MERGE_SORT ($A, mid + 1, r$)] IMPERA

MERGE (A, l, mid, r)] COMBINA

DIVIDE

ET

IMPERA

$l=1$
 $r=9$
 $mid = \frac{9+1}{2} = 5$

$1 \quad 3 \quad 5 \quad 2 \quad 4 \quad 1 \quad 2 \quad 6 \quad 3 \quad 8 \quad | \quad 5 \quad 7 \quad 9 \quad 4 \quad 6 \quad 6^9$

MS(1,9)

$l=1$
 $r=5$
 $mid = \frac{1+5}{2} = 3$

$1 \quad 3 \quad 5 \quad 2 \quad 4 \quad 1 \quad | \quad 2 \quad 6 \quad 3 \quad 8^5$

$l=1$
 $r=3$
 $mid = \frac{1+3}{2} = 2$

$1 \quad 3 \quad 5 \quad 2 \quad | \quad 4 \quad 1^3$

$l=1 \quad r=2$
 $mid = \frac{1+2}{2} = 1$

$1 \quad 3 \quad 5 \quad 2^2$

3

$5 \quad 2 \quad l=2 \quad r=2$

$3 \quad 5 \quad 2$

$merge(A, 1, 1, 2)$

T. Hege

$$T_n(n) = \theta(n) + \theta(n) = \theta(n)$$

$n = n - l + 1$ for 3 while

$$T_{HS}(n) = \begin{cases} \theta(1) & n \leq 1 \text{ base case} \\ T_{HS}(n/2) + T_{HS}(n/2) + \theta(n) = \lg(n) \cdot cn + o(n) = \Theta(n \lg(n)) \end{cases}$$